

Special products



1

a False

$$8^2 + 96 + 6^2$$

b False

$$y^2 - 14y + 49$$

2

a $(x + 3)^2 = x^2 + 6x + 9$

c $x^2 + 64 - 16x = (x - 8)^2$

e $(8x^2 - 9)^2 = 64x^4 - 144x^2 + 81$

g $(2y + 2)(2y - 2) = 4y^2 - 4$

b $(x - 3)^2 = x^2 - 6x + 9$

d $(x + 6)^2 = x^2 + 12x + 36$

f $(-9x + 2)^2 = 81x^2 - 36x + 4$

h $(3x - 8)^2 = 9x^2 - 48x + 64$

3

a $t^2 + 10t + 25$

b $t^2 + 10t + 25$

c

i No ii Yes iii No iv Yes

4

a $x^2 + 6x + 9$

c $s^2 + 8st + 16t^2$

e $81x^2 + 36x + 4$

g $16x^2 + 56xy + 49y^2$

i $81x^2 + 432xy + 576y^2$

k $\frac{64x^2 - 144x + 81}{81}$

m $\frac{n^2}{16} - 2 + \frac{16}{n^2}$

b $m^2 - 10m + 25$

d $x^4 + 6x^2 + 9$

f $16 - 40m + 25m^2$

h $27x^2 + 144xy + 192y^2$

j $147x^3 - 210x^2y + 75xy^2$

l $36y^2 + 6y + \frac{1}{4}$

5

a $m^2 - 9$

b $x^2 - 49$

c $9x^2 - 64$

d $16x^2 - 9y^2$

e $16y^2 - 81$

f $36 - p^2$

g $49y^2 - 64$

h $121 - 81p^2$

i $100 - m^2$

j $384x^2 - 486y^2$

k $20x^3 - 405xy^2$

l $0.64x^2 - 81$

m $81x^2 - 0.16$

n $0.04x^2 - 121$

o $v^2 - \frac{1}{49}$

p $36x^2 - \frac{1}{9}$

q $36x^2 - \frac{64}{9}$

r $x^2 - \frac{4}{x^2}$

s 111



6 Example answer:

$$\begin{aligned}403^2 &= (400 + 3)^2 \\&= 400^2 + 2 \cdot 400 \cdot 3 + 3^2 \\&= 160\,000 + 2400 + 9 \\&= 162\,409\end{aligned}$$

7 Example answer:

$$\begin{aligned}896^2 &= (900 - 4)^2 \\&= 900^2 - 2 \cdot 900 \cdot 4 + 4^2 \\&= 810\,000 - 7200 + 16 \\&= 802\,816\end{aligned}$$

8 Example answer:

$$\begin{aligned}201 \cdot 199 &= (200 + 1)(200 - 1) \\&= 200^2 - 1^2 \\&= 40\,000 - 1 \\&= 39\,999\end{aligned}$$

9

a $16y^2 + 24y + 9 \text{ units}^2$

b $16y^2 + 24y + 9 \text{ units}^2$

10

a $62x^2 - 25y^2$

b $62x^2 - 25y^2$

11

a $6x^2 + 36x + 54 \text{ units}^2$

b $x^3 + 9x^2 + 27x + 27 \text{ units}^3$

12 $m^3 - m$

13

a $y^4 - 81$

b $y^5 - 81y \text{ km}$



Additional questions

14

a $(a + b)^2 = (a + b)(a + b)$
 $= a(a + b) + b(a + b)$
 $= a^2 + ab + ab + b^2$
 $= a^2 + 2ab + b^2$

b $(a - b)^2 = (a - b)(a - b)$
 $= a(a - b) - b(a - b)$
 $= a^2 - ab - ba + b^2$
 $= a^2 - 2ab + b^2$

c $(a + b)(a - b) = a(a - b) + b(a - b)$
 $= a^2 - ab + ba - b^2$
 $= a^2 - b^2$

15

a $(20 + 1)^2 = (20^2 + 2(20) + 1) = 441$

b $(20 - 1)^2 = (20^2 - 2(20) + 1) = 361$

c $(30 + 3)^2 = (30^2 + 2(90) + 9) = 1089$

d $(50 - 3)^2 = (50^2 - 2(150) + 9) = 2209$

16 $4x^2 - 8x + 4 \text{ cm}^2$

17 Rewrite $105^2 - 104^2$ as the difference of squares $(105 + 104)(105 - 104)$ and simplify each binomial individually to get $(209)(1) = 209$

18

a Variable choices will vary.

Let x represent the side length of the square and let y represent the width of the border. Then the area of the square will always be $(x + y)^2 = x^2 + 2xy + y^2$

b Let $x = (2m + 1)$ and let $y = 3$ so that $(x + y)^2$ becomes $(2x + 1 + 3)^2 = (2x + 4)^2$. Then the total area is $4x^2 + 16x + 16$